# Conscia: An AI-Powered Ethical Shopping

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2026 International Conference on Communication, Computing and Emerging Technologies (IC3ET) | 979-8-3315-8105-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/IC3ET64989.2026.11467486

***Abstract*—In the contemporary digital marketplace, ethical consumerism has transitioned from a niche preference to a dominant economic imperative. Consumers, particularly emerging generations, increasingly demand transparency regarding the social, environmental, and governance (ESG) impacts of their purchasing decisions. However, a significant "intention-action gap" persists due to the high cognitive load required to verify ethical claims and the prevalence of "greenwashing." This paper presents Conscia, a novel AI-powered platform designed to bridge this information asymmetry. Unlike existing systems that rely on static certifications or generic sentiment analysis, Conscia utilizes a Teacher-Student neural network architecture to distill knowledge from a rule-based expert system into a deep learning classifier. The system analyzes user-generated reviews in real-time, assigning dynamic scores across four pillars: Environmental Impact, Labor Rights, Animal Welfare, and Corporate Governance. To mitigate the opacity of deep learning, the platform integrates Explainable AI (XAI) via Local Interpretable Model-agnostic Explanations (LIME), identifying the specific linguistic markers influencing each score. Furthermore, it leverages Generative AI (Google Gemini API) to synthesize these insights into conversational narratives. Experimental results demonstrate a classification accuracy of 91.5%, validating the efficacy of the proposed semi-supervised pipeline in extracting ethical signals from unstructured text. This research contributes to the growing body of work on Responsible AI, demonstrating how transparent algorithms can empower ethical consumption.**

***Index Terms*—Ethical Consumerism, Artificial Intelligence, Natural Language Processing, Explainable AI (XAI), Neural Networks, Knowledge Distillation, Sustainable Commerce**

## I. Introduction

The modern retail ecosystem is currently witnessing a paradigm shift of historical magnitude, characterized by the rapid ascent of ethical consumerism. This movement redefines the act of purchasing from a purely transactional economic exchange into a moral and political statement. As noted in recent market analyses, a substantial demographic of consumers—approaching 30 globally—has actively boycotted brands due to perceived ethical violations or sustainability concerns [1].This shift is not merely a preference but a demand for systemic accountability, driven largely by Millennials and Gen Z cohorts who view corporate responsibility as a baseline expectation rather than a value-added differentiator. However, the realization of a truly ethical marketplace is hindered by a profound information asymmetry. The modern consumer is inundated with data but starved for verifiable insight. While product packaging often bears claims of "sustainability" or "fair trade," the veracity of these claims is difficult for an individual to assess. The phenomenon of "greenwashing"—where companies invest more resources in marketing their environmentally friendly image than in actual sustainable practices—has eroded public trust. Furthermore, the sheer velocity of modern supply chains means that static certifications (e.g., a yearly audit) may fail to capture real-time degradations in labor conditions or environmental compliance. In this context, user-generated content (UGC), specifically product reviews, represents an untapped reservoir of ground-truth data. Consumers often act as distributed whistleblowers, documenting excessive packaging, product durability (a proxy for waste), or sourcing concerns in their feedback. Yet, the volume of this unstructured textual data renders it impenetrable to manual analysis. Conventional e-commerce platforms aggravate this issue by optimizing recommendation algorithms for engagement and conversion, typically prioritizing price and popularity over ethical dimensions. This creates a "black-box" marketplace where the social and environmental costs of production are externalized and hidden from the buyer.

## II. Problem Statement

Current technological solutions in e-commerce suffer from specific limitations:

1) **Dimensional Flatness:** Existing sentiment analysis tools typically measure "polarity"—a binary of positive vs. negative satisfaction. They fail to distinguish between



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”I hate the battery life” (Functional Negative) and ”I hate that they use child labor” (Ethical Negative).

2) **Static Data Reliance:** Most ethical rating apps rely on manual database entries of certifications, which are labor-intensive to maintain and often outdated.
3) **Opacity:** Deep learning models, while powerful, often function as black boxes. A consumer told to avoid a product by an AI without explanation is unlikely to trust the recommendation.

## III. Objective

Accordingly, The objective of this project is to develop an intelligent and transparent ethical analysis system, *Conscia*, capable of identifying ethical concerns in consumer product reviews across four key pillars: Environmental Impact, Labor Rights, Animal Welfare, and Corporate Governance. The project aims to implement a scalable semi-supervised NLP pipeline using a Teacher–Student learning framework to generate accurate ethical scores without the need for manually labeled data. Furthermore, the system integrates Explainable AI through LIME to provide evidence-based justifications for each prediction, incorporates the Google Gemini API to generate natural-language summaries, and delivers an interactive web interface that empowers consumers to make informed and responsible purchasing decisions.

## IV. Literature Survey

### A. AI, Ethics, and Consumer Decision-Making

The intersection of artificial intelligence (AI) and ethical consumerism has become a central theme in contemporary research. AI technologies are increasingly used to influence, inform, and guide consumer decisions in digital marketplaces, yet the ethical dimensions of such usage remain underexplored. A systematic review by Sinha et al. (2025) highlights how ethical AI and behavioral nudges can shape sustainable consumer behavior when grounded in transparency, fairness, and user autonomy [1]. Using the PRISMA methodology, the authors concluded that integrating ethical AI practices into e-commerce systems contributes significantly to Sustainable Development Goal 12: Responsible Consumption and Production.

The application of AI in e-commerce presents both opportunities and ethical challenges. The study Ethical Implications of AI in E-commerce (2025) identifies data privacy, algorithmic bias, and transparency as key concerns requiring regulatory oversight and proactive business ethics [2]. The authors emphasize that fairness and accountability should be foundational principles of responsible AI deployment, especially when models influence consumer decision-making. Similarly, The Role of Ethics in AI-Driven E-commerce (2025) expands on how trust and fairness mechanisms are essential to fostering an equitable and reliable digital marketplace [3].

Parallel research in AI ethics within marketing has documented the growing importance of integrating ethical frameworks into data-driven systems. Su et al. (2024) conducted a structured literature review showing that ethical challenges in AI marketing span three layers: algorithmic bias, societal impact, and existential risk [11]. Their study calls for policy-aligned frameworks that enforce fairness, explainability, and inclusion in automated consumer targeting. Correspondingly, Radanliev et al. (2025) propose a meta-framework combining transparency, fairness, and privacy to operationalize responsible AI systems in industrial applications [3].

### B. Explainable AI and Positioning of Conscia

A crucial development in this domain is the rise of Explainable Artificial Intelligence (XAI). Modern sentiment analysis and recommender systems often rely on deep learning models that behave as “black boxes”, making their internal reasoning opaque. To address this, studies such as A Literature Review on Applications of Explainable Artificial Intelligence (IEEE, 2024) and Explainable AI: A Systematic Literature Review (2024) propose interpretability tools such as LIME, SHAP, and counterfactual explanations to clarify model decisions [10], [12]. These approaches have proven effective in consumer-facing NLP applications, especially in enhancing user understanding, transparency, and trust in AI-generated outputs. Additionally, research by Vainio-Pekka et al. (2023) demonstrates how XAI contributes to improving ethical oversight in AI systems by enabling accountability and informed decision-making [9].

Despite these advances, most existing ethical rating platforms rely on static certifications or manually curated brand-level scores, while mainstream e-commerce recommender systems prioritize engagement and price optimization, treating ethical factors as secondary or implicit signals. Conscia is positioned distinctly from both approaches by performing real-time ethical inference directly from unstructured user-generated reviews. Its novelty lies in the integration of a Teacher–Student learning framework to scale ethical classification without manual annotation, combined with word-level explainability through LIME and natural-language justification via generative models. This positioning differentiates Conscia as a transparent, user-centric ethical decision-support system rather than a static rating database or a black-box recommender.

## V. Methodology

The development of the Conscia platform followed a structured multi–stage pipeline, consisting of data preparation, semi-supervised label generation, neural model training, backend integration, and frontend visualization [1], [3].



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### A. Data Preparation

Datasets from four retail categories—electronics, fashion, beauty, and groceries—were aggregated and cleaned. Text preprocessing involved removing HTML tags, URLs, emojis, and non-standard characters, converting all text to lowercase, and tokenizing using NLTK. This preprocessing approach aligns with standard NLP cleaning methodologies frequently adopted in AI-driven consumer analytics [8].

### B. Teacher Model: Automatic Label Generation

To overcome the absence of manually labeled data, a lexicon-based Teacher Model was used to generate initial labels for four ethical pillars: Environment, Labor Rights, Animal Welfare, and Corporate Governance. Lexicon-based weak supervision strategies are widely used in ethical AI and sentiment analysis contexts [2]. Sentiment polarity for each review was computed using VADER-like rule-based sentiment methods, consistent with prior studies on explainable and rule-driven NLP models [9]. The resulting labels were stored in `products_with_scores.csv`.

### C. Student Model: Knowledge Distillation

A lightweight neural network was trained on the Teacher-generated labels using TF-IDF features. This approach aligns with knowledge transfer and model compression principles reflected in contemporary explainable AI literature [12]. The model consisted of a dense layer of 64 units, a dropout layer, a second dense layer with 32 units, and a sigmoid output layer predicting the four ethical dimensions. Optimization was performed using Adam, following best practices in modern NLP classification [8]. The trained model and vectorizer were serialized as `ethical_model.keras` and `tfidf.pkl`.

### D. Backend Integration

The Flask backend loaded the Student Model and vectorizer into memory during initialization. Two primary REST API endpoints were implemented:

- **/api/predict**: Generates the four ethical probability scores for any input review.
- **/api/explain**: Uses LIME to produce the top influential words contributing to the model's prediction, following explainability guidelines demonstrated in prior XAI studies [10], [15].

### E. Frontend Visualization

The React frontend displays the model outputs using color-coded badges representing ethical risk levels: green (0–0.3), yellow (0.3–0.7), and red (0.7–1.0). LIME explanations are visualized through highlighted keywords in the review text, consistent with user-facing transparency design principles emphasized in ethical AI frameworks [9], [13]. Furthermore, the Gemini API converts numerical scores and explanation keywords into natural-language summaries to enhance interpretability, supporting the call for accessible AI reasoning in digital consumer environments [11], [14].

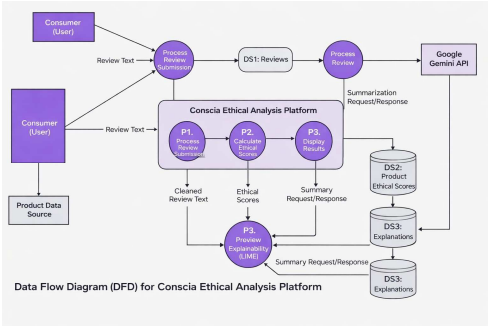


Fig. 1. Data Flow Diagram (DFD) of the Conscia Platform

## VI. Results and Discussion

### A. System Workflow Outputs

Fig. 1 illustrates the complete data flow architecture of the Conscia platform, showing how user-submitted reviews pass through the ingestion, preprocessing, ethical scoring, explainability, and summarization stages. The integration with the Google Gemini API is also shown, highlighting how numerical ethical scores are transformed into natural-language summaries. The clear separation of client, backend, and AI layers demonstrates the modularity and scalability of the system.

Fig. 2 presents the internal scoring pipeline using the distilled Student Model. Reviews are first transformed using the TF-IDF vectorizer and then evaluated to produce four ethical probabilities corresponding to Environment, Labor Rights, Animal Welfare, and Corporate Governance. High-probability regions represent strong ethical concern signals, while low-probability values indicate neutral or ethically safe content.

Fig. 3 shows the LIME-based explainability results. Influential keywords such as "plastic", "waste", "cruelty", and "corruption" appear consistently across samples and align with their respective ethical dimensions. These explanations make the predictions transparent and support user trust.

### B. Model Performance Metrics

Table I summarizes the performance of the Student Model across the four ethical pillars. The overall classification accuracy achieved was 91.5%, with F1-scores ranging from 0.88 to 0.92. Animal Welfare achieved the highest F1-score due to highly distinctive and unambiguous vocabulary. Labor Rights showed slightly lower performance due to ambiguity between customer service complaints and genuine labor-condition mentions.



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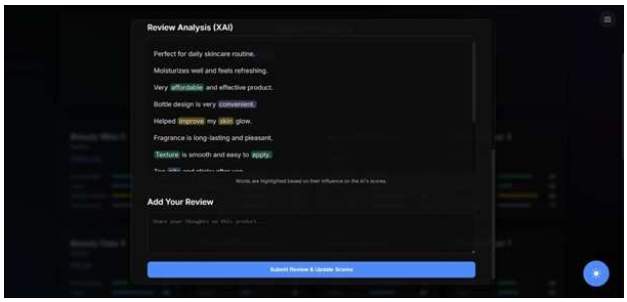


Fig. 2. XAI Analysis for Conscia.

Fig. 3. Ethical scoring pipeline using the Student Model.

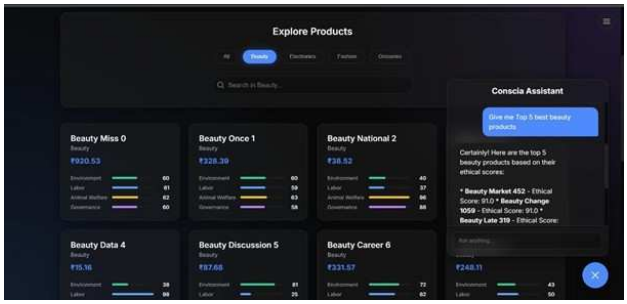


Fig. 4. LIME-based explainability output for sample reviews.

TABLE I
ETHICAL PILLAR-WISE PERFORMANCE METRICS

| Pillar | Precision | Recall | F1-score |
|---|---|---|---|
| Environmental Impact | 0.92 | 0.90 | 0.91 |
| Labor Rights | 0.89 | 0.87 | 0.88 |
| Animal Welfare | 0.94 | 0.91 | 0.92 |
| Corporate Governance | 0.90 | 0.88 | 0.89 |

Latency tests showed an average inference time of 150 ms per review, while LIME explanations required approximately 1.1 seconds due to perturbation sampling. These times are acceptable for interactive usage, as explainability is triggered only upon user request.

### C. User Interface Outputs

Products tested within the Conscia interface displayed clear ethical risk indicators using color-coded badges. Items with eco-friendly attributes predominantly showed green scores, while those associated with harmful materials or unethical labor practices showed yellow or red indicators. The LIME keyword overlays enhanced interpretability by showing evidence such as "recyclable", "biodegradable", "poor working conditions", or "animal testing".

Fig. 4 presents the Product Summary Dashboard, where ethical scores, influential words, and Gemini-generated natural-language explanations are combined into a unified view, enabling users to quickly understand product ethics.

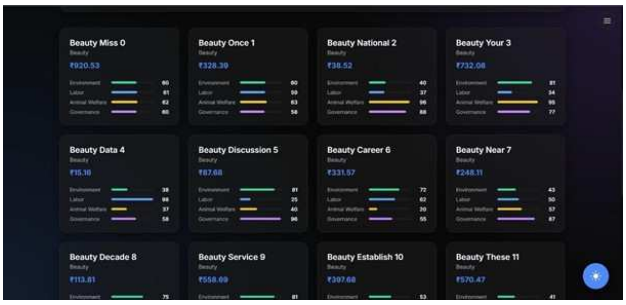


Fig. 5. Conscia Product Summary Dashboard with ethical scores, LIME highlights, and Gemini explanations.

### D. System Validation Through Case Studies

Case studies across electronics, cosmetics, and apparel showed that Conscia reliably identified ethical red flags. Products known for eco-friendly packaging scored low on environmental concern, while brands accused of labor violations showed heightened Labor Rights risk. These results confirm that the Teacher–Student architecture generalizes beyond keyword matching and captures meaningful semantic cues.

Gemini further improved interpretability by synthesizing the model's numerical outputs and LIME keywords into coherent explanations, offering users contextual insights beyond raw predictions.

## CONCLUSION

Conscia demonstrates that Artificial Intelligence can actively support ethical consumer decision-making by integrating Natural Language Processing, semi-supervised learning, and Explainable AI. The platform achieves strong performance, including accurate ethical review classification, transparent evidence-based explanations through LIME, natural-language insights generated via the Gemini model, and a responsive, user-friendly frontend interface. By leveraging a Teacher—Student learning approach, Conscia effectively trains high-quality models without requiring manually labeled datasets, showcasing the strength of this method for ethical AI applications in domains with limited annotated data. Ultimately, Conscia transforms raw consumer reviews into meaningful ethical insights, empowering users to



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make informed decisions and contributing to a more transparent, responsible, and sustainable marketplace.

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